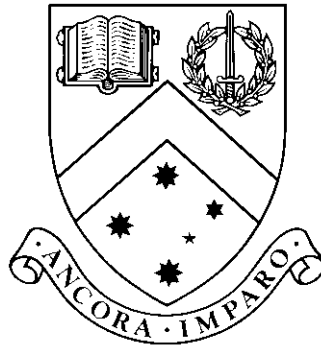


Closed-loop Feedback in Generative Active Learning with Large Language Models

by

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Literature Review

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Chapter 1

Introduction

In the research and application of Machine Learning (ML), the acquisition and annotation cost of data has always been a crucial factor affecting both model performance and its practical deployment. Traditional Supervised Learning usually requires a large amount of labeled data; however, in many real-world scenarios, data annotation is both expensive, time-consuming and high professional demand, such as in speech recognition, medical image diagnosis, legal document classification, or information extraction in highly specialized domains. To reduce the need for extensive annotation while improving training efficiency, Active Learning (AL) has emerged. Its core idea is that if the most valuable subset of data can be selected from a large unlabeled pool for annotation, it is possible to train a more effective model at a lower cost. (Zhang et al.; 2022)

In the traditional AL framework, the main components include an unlabeled pool, a query strategy, an oracle, a labeled pool, an active learner, and a stopping strategy. Through a closed-loop feedback mechanism and the iterative cycle of query–annotate–train, the model can gradually converge and improve its performance. Among query strategies, classical approaches include Uncertainty Sampling (Lewis and Catlett; 1994) and Version Space Reduction (Cohn et al.; 1994). These methods have been extensively validated in the classical AL literature, demonstrating their effectiveness in accelerating model learning and improving accuracy under limited annotation budgets.

Nevertheless, traditional AL still faces significant limitations. First, it is highly dependent on the task and dataset, different tasks often require the design of entirely new query strategies. In addition, the cold-start problem (Yuan et al.; 2020) makes it difficult to select informative samples in the initial training phase, causing slowing model convergence.

With the rise of Large Language Models (LLMs) and Generative AI, Generative Active Learning (Generative AL) has gradually attracted increasing attention. Leveraging the strong in-context learning and few-shot reasoning capabilities of LLMs, Generative AL can intervene in the AL process in novel ways. For example, an LLM can directly serve as a query strategy by using prompts to filter samples from the unlabeled pool; it can also act as an oracle, either independently or in collaboration with humans; or it can indirectly improve sample selection by generating explanations or synthesizing new data (data augmentation).

However, despite these opportunities, Generative AL also faces major challenges. Most methods treat the LLM solely as either the query strategy or the oracle, without incorporating real-time training dynamics of the active learner into the selection process. This creates a disconnect between query decisions and the actual learning needs of the model. Some studies, such as FreeAL (Xiao et al.; 2023), introduce a Strong–Weak Model collaborative mechanism, but their feedback is limited to label quality rather than reflecting the learner’s true learning state. Therefore, an important research gap remains: *how to design*

a closed-loop mechanism that integrates the active learner’s feedback into LLM-based query decision making?

To address this gap, this study aims to explore the design and application of a Feedback Mechanism in Generative AL. We consider two distinct mechanisms: (a) Direct Feedback, in which the training dynamics of the active learner are converted into textual descriptions and incorporated into the LLM’s query process; and (b) Indirect Feedback, where discrepancies between strong and weak models are leveraged to indirectly reflect the state of the learner. By comparing these approaches with traditional baseline methods, we aim to evaluate the effectiveness of feedback mechanisms in improving both efficiency and generalization in AL.

This report aim to provide the literature review of generative active learning with large language models and the research plan for our project. The structure of this report is as follows:

- **Chapter 2** will present the literature review, first covering the core concepts and query strategies of traditional AL, followed by recent developments and limitations in Generative AL, to identify potential areas for improvement.
- **Chapter 3** will synthesize and critically analyze existing studies, highlighting advantages and shortcomings of current approaches, with a focus on uncovering the research gap related to the integration of active learner feedback.
- **Chapter 4** will outline the research project plan, including baseline methods and the design of direct and indirect feedback mechanisms, and the research timeline, aiming to evaluate how such mechanisms can enhance Generative AL performance.
- **Chapter 5** will conclude by summarizing the findings of the literature review and positioning them as the foundation for the subsequent research phases.

Chapter 2

Substantive Literature Review

2.1 Traditional Active Learning

Active learning(AL) (Zhang et al.; 2022) is a subfield of machine learning. Its core concept is that if a small amount of valuable data can be selected for model training, a better performing model can be obtained with less training. Traditional machine learning training often requires thousands of or even millions of labeled data points (Emam et al.; 2021). When annotation is inexpensive, it is often possible to directly label large amounts of data and conduct supervised training. However, in some cases, annotating data is prohibitively expensive (e.g. speech annotation, text translation, information extraction, document classification, and other highly specialized applications). In these cases, annotating large amounts of unlabeled data is clearly unsuitable.

Active learning can query large amounts of unlabeled data, filter out relatively valuable data for annotating to address the high cost of data annotation. A typical active learning system has the following components: an unlabeled pool, a query strategy, an oracle, a labeled pool, an active learner, and a stopping strategy. Active learning uses a pre-set query strategy to select suitable unlabeled data from the unlabeled pool and send it to the oracle for labeling. The oracle, which can be a human oracle (mostly in traditional AL) or an LLM oracle, is primarily responsible for annotating the selected unlabeled data from queries, placing them in the labeled pool, and training the active learner. The active learner is the machine learning model that is being trained and is typically a relatively small task-oriented model. Active learning uses the active learner’s feedback as an important basis for query, and repeats the query, annotation, and training steps. The stopping strategy specifies when this loop should stop. Common stopping strategies include cost, time, and active learner performance.

In traditional AL, the query strategy significantly impacts the performance of the active learner and the speed of training convergence. This is why numerous studies are trying to identify the most valuable data from various perspectives. The following describes several commonly used query strategies in AL.

2.1.1 Uncertainty

The uncertainty sampling method was first proposed by Lewis and Catlett (1994). Its core idea is to instruct the active learner to avoid re-querying confident instances and instead focus on the least uncertain or unconfident unlabeled instances.

Least Confidence

For classification tasks, there are currently many methods for calculating the uncertainty score in an unlabeled pool. The simplest implementation is the minimum confidence

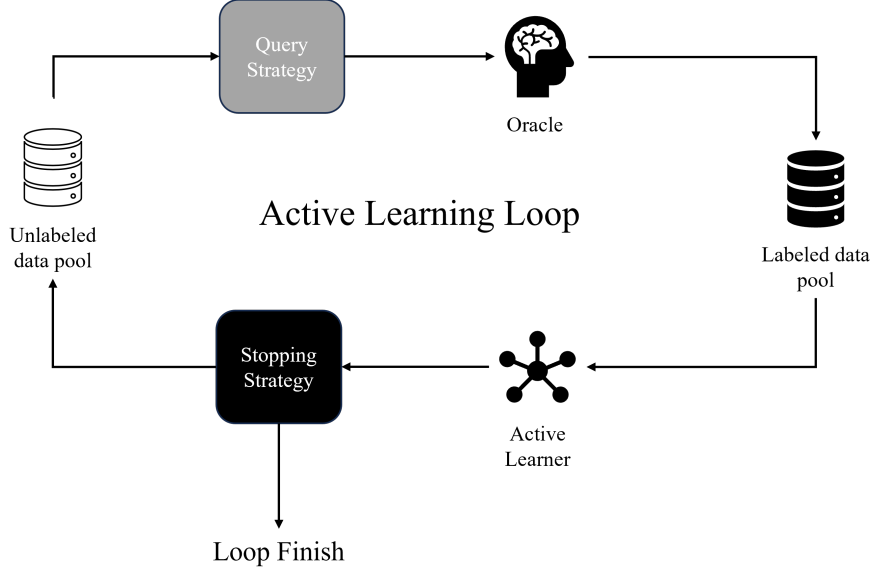


Figure 2.1: The classic AL loop and components. In the AL, it includes an unlabeled pool, a query strategy, an oracle, a labeled pool, an active learner, and a stopping strategy.

method, which directly finds the instance with the lowest confidence based on the model's output probability.

$$x_{LC}^* = \arg \min_x P_\theta(\hat{y} | x) = \arg \max_x (1 - P_\theta(\hat{y} | x)) \quad (2.1)$$

$$\hat{y} = \arg \max_y P_\theta(y | x) \quad (2.2)$$

In equations above, x denotes an unlabeled instance, y denotes a possible class label, $\hat{y}$ is the predicted class with the highest probability, and $P_\theta(y | x)$ represents the predicted probability of assigning label y to instance x under the model parameterized by θ . The least-confidence strategy selects the instance x for which the model is least confident in its prediction. However, this method only considers the information with the highest probability in the prediction of the model and ignores the rest of the prediction distribution.

Entropy

Entropy is a metric that measures the average information contained in a variable and is often considered an uncertainty in machine learning. In AL query strategies, entropy (Shannon; 1948) is a more general strategy for calculating uncertainty scores and is more commonly used.

$$x_e^* = \arg \max_x \left(- \sum_i P(y_i | x; \theta) \log P(y_i | x; \theta) \right) \quad (2.3)$$

Where $P(y_i | x; \theta)$ is the predicted probability of assigning label y_i to instance x under the model parameterized by θ . The entropy-based uncertainty strategy selects the instance with the highest prediction entropy, thus prioritizing samples where the model is most uncertain across the entire probability distribution.

2.1.2 Reducing Version Space

Query by Disagreement

The *version space* (V) (Mitchell; 1977) refers to the set of all hypotheses that are consistent with the current training data, and it gradually shrinks as more data are added. The concept of Reducing Version Space is to select data instances that can maximally reduce the version space. Cohn et al. (1994) proposed the query by disagreement (QBD) method. The core idea is to traverse the version space for new input data in a stream-based scenario. If any two hypotheses disagree with the new data, they are retained; otherwise, they are eliminated. While this method is conceptually simple, the size of the version space can approach infinity as the dimensionality of the hypothesis increases. Specifically, it is almost impossible to store the entire version space in limited memory.

Later, the implicitly instead or most specific/most general hypotheses (Haussler et al.; 1994) methods have been used to address the problem of excessively large version spaces, but these methods still have drawbacks such as efficiency or stability

Query by Committee

The QBD method has two main assumptions:

- QBD primarily measures the entire version space (or an approximate version space).
- QBD focuses solely on binary disagreement and fails to quantify it.

These two assumptions can lead to problems in the real world where the version space is too large and the disagreement measurement is too coarse. In contrast, query by committee (Abe and Mamitsuka; 1998) relaxes these assumptions. For the version space problem, QBC does not use the entire version space. Instead, it samples hypotheses as committees (C) and compares the disagreements within these committees. There are various methods for selecting committee members, such as Bayesian sampling, boosting, and diversity design.

Among the methods for quantifying the degree of disagreement, the first is to measure the uncertainty of the entire committee. The most common method is vote entropy.

$$x_{VE}^* = \arg \max_x - \sum_y \frac{\text{vote}_C(y, x)}{|C|} \log \frac{\text{vote}_C(y, x)}{|C|} \quad (2.4)$$

Where C represents the committee of hypotheses or models. The term $\text{vote}_C(y, x)$ refers to the number of committee members that predict label y for instance x , and $|C|$ is the total number of committee members. The vote entropy strategy measures the disagreement among committee members and selects instances with the highest entropy.

2.1.3 Summary of Traditional AL

The two query strategies commonly used in traditional AL provide a foundation for generative active learning. In fact, there are many other query strategies not mentioned here, such as minimizing expected error (Roy and McCallum; 2001) and density-weighted method (Lang; 1995). However, these strategies have specific requirements for use or data structure and their application scope is not as broad as the methods described here. Reviewing the various traditional query strategies mentioned above, we can clearly see the core of AL: the query strategy is selected based on feedback from the active learner or committee, and these selected strategies are used to train the active learner or committee again, forming a closed loop. This process effectively avoids redundant training data

and reduces model training costs. However, traditional AL also has significant limitations. First, the design of the query strategy is highly dependent on the task or dataset structure, requiring a new query strategy for different tasks or datasets. Furthermore, traditional AL suffers from a cold-start problem (Yuan et al.; 2020), making the effective selection of initial training data a major challenge.

2.2 Generative Active Learning

Generative AI refers to computational techniques that can generate novel and meaningful content (such as text, images, and audio) from training data (Banh and Strobel; 2023). Large language models (LLMs), the most widely used form of generative AI in recent years, offer a wide range of benefits in the field of AL due to their powerful in-context and few-shot reasoning capabilities. Therefore, in this study, we define generative active learning as an AI model learning method that incorporates any LLMs into active learning. In generative AL, the LLM can play two main roles:

- **Query strategy:** Using a LLM as part of or a entire query strategy, through scoring, direct selection, or as a pre-processing model, can provide AL with different perspectives on the query strategy. Some research enhances data by generating explanations or examples or by instantly synthesizing new data to expand the data space.
- **Oracle:** Data obtained from a query can be passed to an oracle to obtain labels. LLMs can serve as an oracle in various studies, complementing or enhancing the annotation results of human oracles, or even completely replacing them. However, when using language models solely as oracles, the issue of annotation noise must be carefully considered.

In addition, other aspects of the AL also change with the inclusion of LLMs. For example, during AL initialization, data instants could be generated using the LLM out of data pool to mitigate AL cold start issues. Alternatively, the stopping strategy of AL could be based on the runtime of LLM cost or time. In the following literature review, we will focus on the application and performance of generative AL in selection.

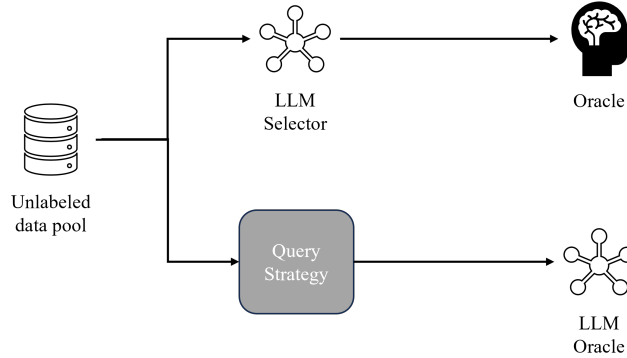


Figure 2.2: LLMs can play the role of query strategy (upper route) and Oracle (lower route) in AL

2.2.1 LLM Query

LLM Selection

Consistent with the goals of traditional AL, generative AL also aims to achieve optimal model performance through query strategy with minimal data annotation costs. The

powerful in-context reasoning and few-shot reasoning capabilities of LLMs allow for greater flexibility in query strategies compared to traditional AL. We define LLM selection as using a LLM for direct selection, selection of eliminated data, or selection by query strategies based on a LLM. Some research has departed from traditional query strategies and has directly employed LLMs for selection. For example, ActiveLLM (Bayer; 2025) prompted an instruction-tuned LLM to filter the unlabeled pool for human annotation, after which a BERT-like model was few-shot trained. By adjusting the prompts, the study significantly improved BERT’s performance on few-shot classification.

Most research, however, leverages the in-context capabilities of LLM to pre-process, transform, or eliminate noisy data within the unlabeled data pool. For example, LDCAL (Li et al.; 2024) prompted an LLM to partition the unlabeled pool into four difficulty levels: Easy, Moderate, Challenge, and Hard. And then applied certainty gain maximization to sample equally from each pool for fine-tuning. This approach significantly improved model stability, effectiveness, and efficiency, outperforming existing methods. ActivePrune (Azeemi et al.; 2024) performs two-stage data pruning using the 5-gram model and a quantized LLM. Then it combines this with traditional NLP acquisition functions for AL. They showed that this method can reduce the end-to-end AL time required for classification tasks by 74%. BAL-PM (Melo et al.; 2024) uses LLM on the human preference alignment task to establish two sets of selecting criteria for AL training. These two selection criteria leverage the embedding capabilities of LLMs and uncertainty measures, taking into account both uncertainty and sample diversity to ultimately obtain the most needed training data.

LLM Generation

Another common LLM query strategy is the LLM generation, in which the LLM generation results are used as the basis for selection. For example, APE (Qian et al.; 2024) uses the active learner as a query strategy, requiring it to answer all unlabeled data k times for each iteration query. The entropy is calculated on the basis of the accuracy of its answers as the selection basis, and the data with the highest entropy are finally selected for few-shot training. Active-Prompt (Diao et al.; 2023) also uses a similar technique. They use LLM to answer multiple questions in a single training session and combine traditional query strategy techniques such as disagreement or entropy to select the data with the highest uncertainty for Chain-of-Thought (CoT) annotation to improve the LLM effect.

Hybrid

There are also various other studies that combine the query strategies of the two LLMs mentioned above. NoiseAL (Yuan et al.; 2025) uses a strong and weak model to jointly train on labeled data, achieving stable training results in classification tasks on noisy datasets. They first divide all data into a clean set R , a hard set H , and a purified set P based on the results of a strong model and a weak model. They then select the top n samples in R that are most similar to P and use LLM to re-label P . Finally, they retrain both models using the re-labeled dataset. CAL (Du et al.; 2024) draws on the sampling concept of AL. In the debias task, they first use llama2 or vicuna to find counter example pairs. After clustering, they use GPT-4 to summarize different types of bias. This study indicated how LLM can provide information for selection and summarize bias type for active learner. Beyond-Labels (Yao et al.; 2023) emphasizes the contribution of explanations to AL sampling. They first calculate the similarity between the explanations in the labeled data pool and the unlabeled data pool, and then perform evenly spaced sampling based on the similarity. This allows the sampling strategy to consider both uncertainty and diversity. They also train an explanation generation model during the

training process to address the problem of unlabeled explanations in inferencing scenarios. FreeAL (Xiao et al.; 2023) is a novel AL architecture that uses a strong and weak model for joint training. The strong model is responsible for labeling the data, while the weak model is responsible for learning and distinguishing noise labels, allowing the training process to completely eliminate the need for human oracles.

Based on the literature in this section, we can see that the core of most generative AL query strategies is still uncertainty-based or disagreement-based methods. The only difference is that LLMs can provide semantic reasoning or semantic embeddings, allowing query strategies to approach and select language tasks from a wider range of perspectives. However, some research uses non-open-source LLM APIs as part of the query strategy, which may make AL results heavily dependent on the capabilities of these LLMs.

2.2.2 LLM Oracle

Many studies have also focused on using LLMs as oracles to reduce human annotation costs. Similar to LLM query strategies, LLMs can also serve as a complete oracle or collaborate with human oracles through a router mechanism. AL-Loop (Kholodna et al.; 2024) was the first to explore the effectiveness of LLMs in labeling low-resource languages, using LLM models as oracles to label data for named entity recognition (NER) tasks. Ultimately, experiments using LLM oracle significantly reduced the cost. Regarding the collaboration between LLMs and human oracles, AutoLabel (Ming et al.; 2024) uses clustering combined with systematic sampling to select the most representative samples and assign them to human experts for labeling. In addition, it incorporates a human feedback mechanism to allow samples that LLM is unsure about to be reassigned to humans for labeling. A similar approach emerged in HybridAL (Rouzegar and Makrehchi; 2024). This study explored how to leverage the confidence of an LLM oracle as a router to determine whether annotation should be performed by humans or LLMs. Ultimately, they found that setting a confidence threshold of 0.8-0.9 allowed the quality of the LLM annotation to approach that of human annotation, significantly reducing costs. Both studies demonstrated the ability of the LLM oracle to significantly reduce the cost of human annotation. In LLMAAA (Zhang et al.; 2023), they employed a traditional AL query strategy with a stronger Teacher LLM for annotation, and trained a weaker Student LLM through robust training to enhance the model’s performance under noisy label conditions.

2.2.3 Limitations of Generative AL

Looking at the generative AL methods described above, we can see that the incorporation of LLM has indeed significantly transformed AL approaches to language or text tasks. However, we also observe some limitations. First, when combining a language model with a traditional AL query strategy as a criterion for data selection (e.g., LDCAL), while we can assess and obtain the learning status of the active learner to a certain extent as feedback and query basis. However, AL training may suffer from a significant cold-start effect in the early stages. Furthermore, as the task type changes, different query strategies must be designed to address each task.

In contrast, the powerful few-shot reasoning capabilities of LLM can significantly reduce the cold-start effect and improve task generalization. However, when LLM is used as a query strategy, it cannot receive the learning status of the active learner as the query criteria. In this case, the active learner can only passively receive and learn on the training data. FreeAL (Xiao et al.; 2023) uses the active learner to feedback the status of the label to improve this situation, but in this research, feedback of the active learner only reflects the quality of the label, not the learning status. Existing research has rarely considered enabling the active learner to provide feedback on its learning state for integration into

the query strategy in the generative AL field. This creates a research gap: how to effectively integrate the immediate learning state of the active learner into the generative AL selection mechanism.

We believe that designing a framework that captures the active learner’s training dynamics and provides structured feedback to the LLM will further enhance the example selection performance of generative AL. Such a mechanism would not only avoid the disconnect between the LLM query strategy and the actual needs of the active learner, but also establish an AL framework that is closer to closed-loop feedback, enabling more efficient and adaptable example selection strategies for diverse tasks.

Chapter 3

Summary of the State of the Art

Traditional Active Learning (AL) centers on maximizing performance under a fixed annotation budget, relying on query strategies to extract high-value samples from an unlabeled pool. Classical strategies include uncertainty-based methods and version space reduction. These approaches demonstrate that when queries align with the immediate training dynamics of the active learner, models can converge with fewer labeled samples. However, they also face limitations such as cold-start issues, strong model or task dependency, and difficulties in accurately estimating uncertainty in high-dimensional semantic tasks.

With the incorporation of LLM, Generative AL has reshaped the landscape of sample selection and annotation. As a query strategy, an LLM can directly perform LLM selection such as ActiveLLM (Bayer; 2025), or serve as a pre-processor or filter in combination with traditional strategies such as LDCAL (Li et al.; 2024), ActivePrune (Azeemi et al.; 2024), and BAL-PM (Melo et al.; 2024). Other approaches use LLM generation to drive queries such as APE (Qian et al.; 2024) and Active-Prompt (Diao et al.; 2023). Hybrid methods also exist, such as NoiseAL (Yuan et al.; 2025) and FreeAL (Xiao et al.; 2023) with strong-weak model co-training, CAL (Du et al.; 2024) with bias summarization with LLMs, and Beyond-Labels (Yao et al.; 2023) with explanation similarity-guided sampling. As an Oracle, LLMs have been used in AL-Loop (Kholodna et al.; 2024), AutoLabel (Ming et al.; 2024), HybridAL (Rouzegar and Makrehchi; 2024), and LLMAAA (Zhang et al.; 2023), where they collaborate with human oracles to reduce costs, though introducing annotation noise and dependence on LLM capability.

Overall, Generative AL brings three major advances to the AL field:

- Few-shot reasoning that mitigates cold-start issues and improves task transferability;
- Semantic-level understanding and generation that enhance flexibility in sample selection;
- Natural language rationales that strengthen interpretability.

However, clear limitations remain. First, most methods position the LLM either as the query strategy or the oracle, without systematically incorporating the real-time training state of the active learner into the LLM’s selection decisions. Some exceptions, such as FreeAL, include label quality feedback, but not actual training dynamics of the learner, preventing the establishment of a true closed-loop feedback mechanism. Second, when tasks change, new strategies often must be manually designed, wasting the task generalization advantages that LLMs are expected to provide.

To date, no systematic methodology or evaluation framework exists for integrating active learner training dynamics into generative AL selection. The absence of such feedback mechanisms risks a disconnect between the LLM query strategy and actual needs of the learner, limiting sample efficiency and cross-task generalization. We propose establishing

a closed-loop mechanism, where the active learner proactively provides feedback to the LLM query strategy. By designing structured prompts/contexts digestible by the LLM, the query strategy can adaptively choose samples based on needs of the learner. This direction can simultaneously alleviate cold-start effects, reduce task-specific customization costs, and provide a reproducible and extensible foundation for Generative AL.

Chapter 4

Research Project Plan

The literature review reveals that the feedback mechanisms in LLM-based generative AL is under-explored. This leads us to two main research questions:

1. How to incorporate feedback mechanism to generative AL?
2. What feedback mechanism is most effective for language models in generative AL?

In our study, we aim to design two different feedback mechanisms and compare them with baseline methods to fully address these two research questions.

4.1 Baseline & Datasets

Among the baseline methods, we will adopt LLM selection methods from generative AL that do not provide feedback. Methods such as ActiveLLM (Bayer; 2025) directly use prompts to instruct the selection model to sample. The prompts are expected to include information such as the task description, guidelines, the number of samples to be sampled, and whether chain of thought (CoT) is required, but do not include active learner feedback. On the task side, we aim to test the generalization ability of the feedback mechanism across a wider range of tasks. We hope to include tasks such as Entity Matching, Classification, Reasoning, Natural Language Inference (NLI), and compare the performance of methods with and without feedback mechanisms on these tasks.

4.2 Feedback Mechanism

Due to the structure of the LLM, without multimodality enabled, the language model cannot directly be fed structured data. Therefore, the most straightforward approach is to feed training dynamics of the active learner into the LLM query strategy via text for next-round sample selection. In this approach, we will focus on prompt design and how to convert training dynamics of the active learner into valid text and integrate it into the prompt.

In addition, inspired by the strong-weak model mechanisms in NoiseAL (Yuan et al.; 2025) and FreeAL (Xiao et al.; 2023), we propose to further integrate such mechanisms into both the LLM query strategy and the active learner. In this scenario, instead of having the active learner directly generate feedback, feedback is derived indirectly through the learning discrepancies between the strong and weak models. Strong models have better reasoning and memory capabilities, while weak models have the opposite. We can evaluate and transform the performance of two models under the same labeled data pool to generate indirect feedback. At the same time, the query strategy must also monitor the changes in

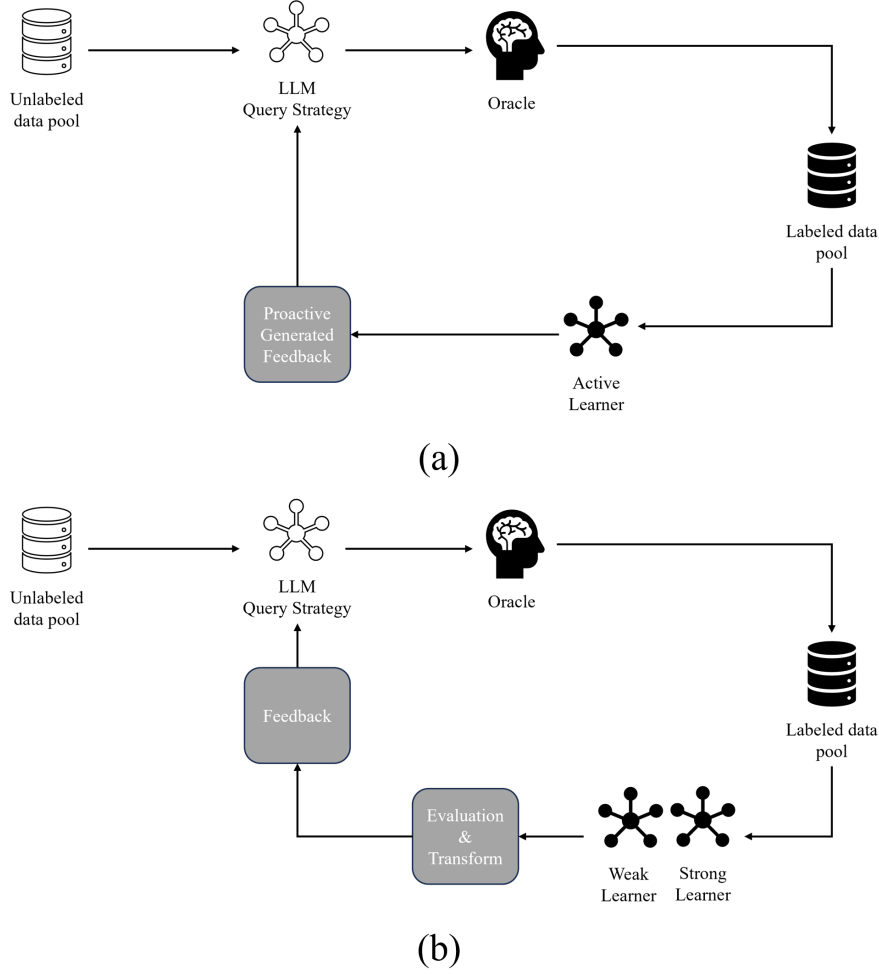


Figure 4.1: Schematic illustration of feedback mechanisms: (a) direct feedback mechanism, and (b) indirect feedback mechanism.

the active learner’s feedback across iterations to ensure that sample selection remains as close as possible to the actual needs of the active learner.

4.3 Timeline

In this study, we plan to spend one month preparing the dataset and establishing the baseline method, the next two months designing and implementing the feedback mechanism, and the remaining time focusing on evaluation and writing.

The following is a detailed timeline:

- Sep 2025-Oct 2025 Datasets and Baseline method
- Nov 2025-Dec 2025 Direct Feedback Mechanism
- Jan 2026-Feb 2026 Indirect Feedback Mechanism
- Mar 2026-Apr 2026 Evaluation
- May 2026-Jun 2026 Thesis Writing

Chapter 5

Conclusion

This study reviewed the core methods and limitations of both traditional Active Learning (AL) and Generative Active Learning (Generative AL). While traditional AL can effectively reduce annotation requirements, it is constrained by the cold-start problem and strong task dependency. In contrast, Generative AL leverages the semantic reasoning and generative capabilities of Large Language Models (LLMs), demonstrating greater flexibility and cross-task potential. However, most existing Generative AL approaches treat LLMs solely as query strategies or oracles, without incorporating the real-time training dynamics of the active learner into decision making. This results in a gap between query decisions and the actual learning needs of the model.

We considered two feedback mechanisms to address this gap: Direct Feedback and Indirect Feedback. Direct feedback integrates the training dynamics of the active learner into the LLM’s query process through textual prompts, while indirect feedback leverages the discrepancies between strong and weak models to indirectly reflect the learner’s state. Both methods are expected to make query strategies more aligned with learning needs, establishing a closed-loop feedback mechanism in AL framework that improves query strategy efficiency and cross-task adaptability.

Through this literature review, we have examined the evolution of AL from traditional frameworks to Generative AL. By comparing their respective methodologies, advantages, and limitations, the review highlights how the integration of LLMs has improved the AL. More importantly, it clarifies the specific gaps that persist within the field, thereby providing a structured foundation for the subsequent research design and analysis.

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